

New body parts

Today, there are many different artificial body parts, known as prostheses, available. These include devices to help people hear and see better, prosthetic limbs that can move naturally, and even artificial hearts. The oldest known prosthetic body part is a wood and leather toe that was found on an Egyptian mummy dating back to around 800 BCE.

Eyeglasses appear in this 14th century German painting



Seeing clearly

- **What?** Eyeglasses
- **Who?** Roger Bacon; Alessandro di Spina
- **Where and when?** UK, 13th century; Italy, 14th century

The origin of eyeglasses is not at all clear. In the 13th century, the English friar Roger Bacon described using lenses to aid sight. The Italian monks Alessandro di Spina and Salvino degli Armati also developed the idea, with eyeglasses appearing early in the 14th century. The earliest eyeglasses had thick lenses made of quartz.

Contact lenses

- **What?** Soft contact lenses
- **Who?** Otto Wichterle
- **Where and when?** Czechoslovakia, 1961

The first contact lenses were created by the German ophthalmologist Adolf Gaston Eugen Fick in 1888. Made of blown glass, they could be worn for just a couple of hours. Lighter plastic versions were

Hearing better

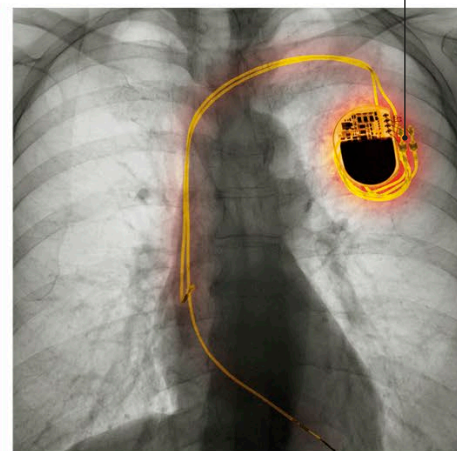
- **What?** Electric hearing aid
- **Who?** Miller Reese Hutchison
- **Where and when?** US, 1898

The first electrical hearing aid came out in 1898, but was quite bulky and had a separate battery pack. The introduction of transistors in the 1950s led to their size being reduced to a device small enough to fit around the ear. Hearing aids got even smaller following the arrival of digital technology in the 1980s.

Hearing aid, 1929



X-ray shows the position of an implanted pacemaker.



Beating hearts

- **What?** Implantable pacemaker
- **Who?** Rune Elmqvist
- **Where and when?** Sweden, 1958

Pacemakers are devices that send small electrical signals to the heart to keep it beating properly. The first pacemakers sat outside the body and greatly limited the patient's movements. Invented in 1958, a pacemaker that is placed inside the body (above) allows patients to live a more normal life.

